

SUMS OF POWERS VIA CENTRAL FINITE DIFFERENCES AND NEWTON'S FORMULA

PETRO KOLOSOV

ABSTRACT. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

CONTENTS

1. Auxiliary lemmas	2
2. Introduction	2
3. Main results	2
4. Conclusions	2
Mathematica programs	3

Date: June 30, 2026.

2010 *Mathematics Subject Classification.* 05A19, 05A10, 11B73, 11B83.

Key words and phrases. Sums of powers, Newton's interpolation formula, Finite differences, Binomial coefficients, Faulhaber's formula, Bernoulli numbers, Bernoulli polynomials, Interpolation, Discrete convolution, Combinatorics, Polynomial identities, Central factorial numbers, Stirling numbers, Eulerian numbers, Worpitzky identity, Pascal's triangle, OEIS.

DOI: <https://doi.org/10.5281/zenodo.18096789>

1. AUXILIARY LEMMAS

In this section, allow us to define a few lemmas, definitions, and conventions that serve as foundation for main results of this manuscript. For multifold sums of powers, we use the notation provided by Donald Knuth in [?], because it is simple and beautiful

Definition 1.1 (Multifold sums of powers). *For non-negative integers r, n, m*

$$\Sigma^0 n^m = n^m,$$

$$\Sigma^1 n^m = \Sigma^0 1^m + \Sigma^0 2^m + \cdots + \Sigma^0 n^m,$$

$$\Sigma^{r+1} n^m = \Sigma^r 1^m + \Sigma^r 2^m + \cdots + \Sigma^r n^m.$$

Lemma 1.2 (Upper Binomial negation). *For integers n, k*

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}$$

Proof. See [?, p. 164, eq. (5.14)]. □

Convention 1.3. *For all x*

$$x^0 = 1.$$

Donald Knuth give extensive discussion on the convention $x^0 = 1$ for all x , including zero, in Concrete Mathematics [?, p. 162], and in [?].

2. INTRODUCTION

3. MAIN RESULTS

4. CONCLUSIONS

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

MATHEMATICA PROGRAMS

Use this [GitHub](#) repository to validate the results using Mathematica programs.

Mathematica Function	Validates / Prints
<code>ValidateBackwardNewtonsFormulaForPowers.txt</code>	Validates Lem. (??)

Metadata

- **Initial release date:** January 3, 2026.
- **Current release date:** June 30, 2026.
- **Version:** Local-0.1.0
- **MSC2010:** 05A19, 05A10, 11B73, 11B83 .
- **Keywords:** Sums of powers, Newton's interpolation formula, Finite differences, Binomial coefficients, Faulhaber's formula, Bernoulli numbers, Bernoulli polynomials, Interpolation, Discrete convolution, Combinatorics, Polynomial identities, Central factorial numbers, Stirling numbers, Eulerian numbers, Worpitzky identity, Pascal's triangle, OEIS .
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- **DOI:** <https://doi.org/10.5281/zenodo.18096789>
- **Web Version:** kolosovpetro.github.io/sums-of-powers-central-differences/
- **Sources:** [github/kolosovpetro/SumsOfPowersCentralDifferencesNewtonFormula](https://github.com/kolosovpetro/SumsOfPowersCentralDifferencesNewtonFormula)
- **ORCID:** [0000-0002-6544-8880](#)
- **Email:** kolosovp94@gmail.com

DEVOPS ENGINEER

Email address: kolosovp94@gmail.com

URL: <https://kolosovpetro.github.io>